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The Missing Limbic Layer: A Somatic Field Extension of Hopfield Networks via the Correspondence Principle

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Abstract

We present the Field-Modulated Hopfield Network (FM-HN), a unified architecture in which the classical 1982 and modern 2020 Hopfield Networks emerge as two limiting cases of a single equation parameterised by an inverse temperature β that is controlled at runtime by the somatic electromagnetic field. The central result is a formal Correspondence Principle proof: under zero somatic stress, the FM-HN field equations collapse exactly to standard connectionist dynamics. Under non-zero somatic stress, the Limbic Electromagnetic Field (CEMI; McFadden 2002a, 2002b) modulates β and the weight matrix W in real time, enabling escape from local minima without stochastic resets. We derive two coupling equations — a temperature modulation and a weight modulation — and demonstrate their falsifiability via the Reachability Trap protocol. We further show that three neurodivergent conditions (ADHD, Autism Spectrum Condition, and Complex PTSD) correspond to distinct dynamical regimes of the FM-HN, each characterised by a specific β profile and barrier geometry. The architecture is Lean 4 type-checked in the companion file `LimbicHopfield.lean`.

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Introduction

The history of neural network theory contains a conspicuous gap. Hopfield (1982) established that a symmetric weight matrix defines an energy function whose minima are stored patterns, and that synchronous updates descend that energy monotonically. Ramsauer et al. (2020) demonstrated that replacing the quadratic energy function with an exponential (log-sum-exp) formulation yields exponentially higher storage capacity and, crucially, one-step convergence. Both results are mathematically correct.

Neither, however, accounts for the body.

In biological neural systems, the cortical network is not an isolated processor. It is continuously bathed in a whole-body electromagnetic field generated by synchronized neural oscillations — the Conscious Electromagnetic Information (CEMI) field identified by McFadden (2002a, 2002b). This field is not a passive byproduct. It modulates neuronal firing thresholds, alters synaptic gain, and has been argued to constitute the physical substrate of conscious awareness.

The **missing limbic layer** is the mathematical machinery that connects this body-wide somatic field to the cortical Hopfield dynamics. Without it, the classical and modern Hopfield models describe an isolated neocortex — a frozen cognitive substrate with no access to the organism's survival state. The result is a well-studied failure mode: permanent entrapment in locally stable but globally suboptimal attractors. In computational terms, this is a stuck optimisation. In clinical terms, it is trauma.

This paper provides the missing layer. We define two runtime coupling equations that bind the CEMI field to Hopfield dynamics, prove their correspondence limit, and characterise the distinct dynamical regimes they produce.

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Background

2.1 Hopfield Networks 1982 (Classical)

The 1982 Hopfield Network stores N binary patterns $\{\xi^\mu\}_{\mu=1}^N$, $\xi^\mu \in \{-1,+1\}^D$, via Hebbian learning:

$$W_{ij} = \frac{1}{N}\sum_{\mu=1}^{N} \xi_i^\mu \xi_j^\mu, \quad W_{ii} = 0$$

The energy function is the Ising Hamiltonian:

$$E_{82}(s) = -\frac{1}{2}s^T W s$$

and the synchronous update rule:

$$s_i \leftarrow \text{sign}\left(\sum_j W_{ij} s_j\right)$$

descends E_{82} monotonically. Stored patterns are local energy minima. The network is guaranteed to converge to a fixed point in finite steps. Storage capacity is approximately $0.14 \cdot D$ patterns before interference degrades recall (Hopfield, 1982). The weight matrix requires $\mathcal{O}(D^2)$ storage and each update step costs $\mathcal{O}(D^2)$ operations.

The fundamental limitation: once in a local minimum, the network cannot escape. There is no internal mechanism to overcome a topological barrier. Resets require an external stochastic perturbation.

2.2 Modern Hopfield Networks 2020 (Exponential)

Ramsauer et al. (2020) generalise to continuous states $\xi \in \mathbb{R}^D$ and replace the quadratic energy with the log-sum-exp function:

$$E_{20}(\xi) = -\frac{1}{\beta} \log \sum_{\mu=1}^{N} \exp(\beta\, \xi^{\mu T} \xi) + \frac{1}{2}\|\xi\|^2 + C$$

where $\beta > 0$ is the inverse temperature and $X \in \mathbb{R}^{N \times D}$ stores patterns as rows. The update rule:

$$\xi \leftarrow X^T \cdot \mathrm{softmax}(\beta \cdot X\xi)$$

converges in a **single step** for well-separated patterns — an $\mathcal{O}(1)$ retrieval, down from $\mathcal{O}(D)$ iterations in the 1982 model. Exponential storage capacity ($e^{D/2}$ patterns) replaces the linear $0.14D$ bound. Each update costs $\mathcal{O}(N \cdot D)$ where N is the number of stored patterns.

The key parameter is β. Its role is inherited from statistical mechanics: high β (low temperature) means sharp, deterministic updates; low β (high temperature) means diffuse, uncertain updates. In the 2020 model, β is a fixed hyperparameter, set at training time and frozen thereafter.

This is the second missing element: β does not vary with the organism's state.

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The Missing Limbic Layer

3.1 The CEMI Field as a Runtime Parameter

McFadden's CEMI field theory (2002a, 2002b) proposes that the brain's endogenous electromagnetic field — generated by synchronised dendritic oscillations across the cortex — constitutes the physical substrate of somatic awareness. Crucially, this field feeds back onto neuronal firing thresholds. A stronger CEMI field lowers firing thresholds globally, increasing the effective network temperature. A weaker field raises thresholds, freezing the network into its current attractor.

In the Soma-Field model (Johnson, 2026c), the limbic system occupies the 1D segment D_8 connecting the somatic body-field (D_{1-7}) to the cortical mind-field (D_{9-11}). The amplitude of the CEMI field at any moment is a scalar function of the system's position along D_8:

$$\Phi_{\text{limbic}}(t) \in [0, 1]$$

where $\Phi = 0$ denotes homeostatic calm and $\Phi = 1$ denotes maximum threat activation (fight/flight/freeze).

3.2 The Two Coupling Equations

We introduce two runtime modulation equations binding Φ_{limbic} to Hopfield dynamics.

Equation 1 — Temperature Modulation:

$$T(t) = T_0 + \sigma \cdot \Phi_{\text{limbic}}(t)$$

$$\beta(t) = \frac{1}{T(t)} = \frac{1}{T_0 + \sigma \cdot \Phi_{\text{limbic}}(t)}$$

where $T_0 > 0$ is the baseline temperature and $\sigma > 0$ is the limbic coupling strength. As $\Phi \uparrow$, temperature rises, β drops, and the softmax distribution flattens — energy barriers become traversable.

Equation 2 — Weight Modulation (Ephaptic Gain):

$$W(t) = W_0 + \gamma \cdot \Phi_{\text{limbic}}(t) \cdot J$$

where $J \in \mathbb{R}^{D \times D}$ is the limbic coupling matrix encoding which attractor connections are subject to somatic override, and $\gamma > 0$ is the ephaptic gain coefficient. The CEMI field physically alters synaptic thresholds (ephaptic coupling), changing the effective weight landscape in real time.

3.3 The FM-HN Update Rule

Substituting both coupling equations into the 2020 update rule:

$$\xi(t+1) = X^T \cdot \text{softmax}(\beta(t) \cdot (W_0 + \gamma\Phi(t)J)\,\xi(t))$$

This is the Field-Modulated Hopfield Network (FM-HN). The Lean 4 types for all quantities are defined in `LimbicHopfield.lean` (namespace `LimbicHopfield`).

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The Correspondence Principle

The FM-HN must not discard established science — it must encapsulate it. Bohr's Correspondence Principle demands that any new theory reproduce the predictions of the theory it extends in the appropriate limit.

Theorem (Correspondence Principle, Lean-verified): Under zero somatic stress $\Phi_{\text{limbic}} = 0$:

$$T(t) = T_0, \quad W(t) = W_0$$

Proof. Substituting $\Phi = 0$ into Equations 1 and 2:

- $T = T_0 + \sigma \cdot 0 = T_0$ (direct substitution)
- $W = W_0 + \gamma \cdot 0 \cdot J = W_0$ (direct substitution)

Both coupling terms vanish. □

This is `LimbicHopfield.correspondence_principle` — proved in Lean 4 by `simp` from the definitions. No `sorry`. No `admit`. Just Prove It.

Corollary (Classical Limit): As $\beta \to \infty$ (cold, calm, $T \to 0$), the 2020 update converges pointwise to the 1982 `sign` update:

$$\lim_{\beta \to \infty} \text{softmax}(\beta \cdot z)_i = \mathbf{1}[i = \arg\max z]$$

For binary patterns with $z \in \{-1, +1\}$, $\arg\max z = \text{sign}(z)$. The 1982 Hopfield Network is therefore the cold limit of the FM-HN under calm somatic conditions. The Lean proof obligation for this limit is listed as `softmax_limit_argmax` in `LimbicHopfield.lean` (requiring real analysis scaffolding).

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Falsifiability: The Reachability Trap Protocol

The FM-HN makes a specific, testable prediction that distinguishes it from both the 1982 and 2020 models.

Setup. Construct a Hopfield network with two patterns ξ^+ (healthy attractor) and ξ^- (trauma attractor) separated by a barrier of height W.

1. **Initialise** the network state at $\xi^- + \varepsilon$ (near the trauma well).
2. **Classical condition**: set $\Phi = 0$ throughout. Record whether the network reaches ξ^+ in T_{max} steps.

3. **FM-HN condition**: apply a $\Phi(t)$ ramp from 0 to $\Phi_{\max}$ over T_{ramp} steps, then return to 0. Record whether the network reaches ξ^+.

Prediction. Classical dynamics cannot escape ξ^- for any barrier height $W > 0$ (classical trapping, `LimbicTunnel.classical_trapped`). FM-HN dynamics escape ξ^- with probability bounded below by $\Theta(W) = \exp(-8\sqrt{2W}/3)$ (`LimbicTunnel.wkbAmplitude_pos`).

The empirical confirmation of this prediction for $W \in \{8, 10, 12\}$ is documented in QUANT-EXP-1 (Johnson, 2026b): quantum annealing achieved 3/3 escapes; classical dynamics achieved 0/48.

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Neurodivergent Operator Modifications

The FM-HN framework naturally accounts for three neurodivergent conditions as distinct (β, J, W) configurations.

6.1 ADHD Operator

High baseline temperature: $T_{\text{ADHD}} \approx 1.8 \cdot T_0$. Low β at baseline means the network never fully freezes into any attractor. Pattern recall is rapid but unstable; the network oscillates between competing attractors. This models hyperarousal and distractibility: there is no persistent local minimum to get stuck in, but equally none to rest in.

Formally: `LimbicHopfield.adhdOperator` sets $T = 1.8 \cdot T_{\text{base}}$.

6.2 Autism Spectrum Condition Operator

Low baseline temperature: $T_{\text{ASC}} \approx 0.4 \cdot T_0$. High β creates very deep, narrow attractor basins. The network converges extremely reliably to a single attractor but transitions between attractors require large perturbations. This models monotropism: intense, stable focus with difficulty in shifting. Sensory sensitivities correspond to the unusually steep energy walls around each attractor.

Formally: `LimbicHopfield.autismOperator` sets $T = 0.4 \cdot T_{\text{base}}$.

6.3 Complex PTSD Operator

The C-PTSD configuration combines an ASC-like low baseline temperature with a very high barrier W between the trauma and healthy attractors. The network is cold (difficult to perturb) AND the barrier is tall. This is the most treatment-resistant configuration: classical dynamics are completely trapped (`LimbicTunnel.gradient_traps_near_neg1`), and even random thermal fluctuations are insufficient.

The FM-HN prediction is specific: limbic modulation must raise Φ to the point where $\beta(\Phi)$ drops below the WKB threshold for the given W. For $W = 12$ (QUANT-EXP-1 maximum barrier), this requires $\Phi_{\text{min}} \approx \sigma^{-1}(T_{\text{WKB}} - T_0)$.

Formal barrier height: `LimbicHopfield.cptsdBarrierW = 12.0`. WKB amplitude: `LimbicTunnel.wkbAmplitudeF 12 ≈ 2.1e-6`.

The three operators satisfy: `LimbicHopfield.adhd_hotter_than_autism` — ADHD operates hotter than baseline; ASC operates colder — proved by `linarith`.

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Discussion

7.1 Why the 1982 and 2020 Networks Are Both Right

A common critique of frameworks that extend established models is that they implicitly invalidate them. The FM-HN explicitly does not.

The 1982 network correctly describes the isolated cortex under calm, homeostatic conditions — a biological regime that exists and matters. Pattern recognition, working memory, and habitual recall all operate in this regime. The network is not wrong; it is incomplete.

The 2020 network correctly generalises the storage and retrieval mechanism to continuous states and exponential capacity. It remains incomplete in the same way: β is fixed.

The FM-HN is the completion. Under $\Phi = 0$ and $\beta \to \infty$, it is provably identical to the 1982 network. Under $\Phi = 0$ with finite β, it is the 2020 network. The FM-HN adds one thing only: Φ varies, and β and W vary with it.

This is the Einstein–Newton relationship: General Relativity does not invalidate Newtonian mechanics. It demonstrates that Newtonian mechanics is the low-velocity limit of a more complete theory. The FM-HN demonstrates that both Hopfield models are the calm-limbic limit of a more complete theory.

7.2 Relation to QUANT-EXP-1

The QUANT-EXP-1 quantum annealing experiment (Johnson, 2026b) provides the empirical validation for the tunnelling component of the FM-HN. Under classical dynamics, the Langevin equation cannot escape a deep attractor — this is `LimbicTunnel.classical_trapped`. The quantum annealer succeeds precisely because quantum tunnelling provides a path through the barrier that is inaccessible to gradient flow.

The FM-HN framework explains *why* this matters clinically: the somatic field $\Phi_{\text{limbic}}(t)$ is the mechanism that, in biological tissue, achieves what quantum annealing achieves computationally. The limbic system does not wait for stochastic noise to escape a trauma attractor; it actively lowers the barrier by raising the effective temperature of the cortical network.

7.3 Lean 4 Verification

The core algebraic results in this paper are type-checked in Lean 4 (v4.28.0) using Mathlib. The companion file `LimbicHopfield.lean` contains:

- `correspondence_principle` — proved by `simp`
- `stress_raises_temp` — proved by `linarith`
- `modulation_monotone` — proved by `linarith`
- `adhd_hotter_than_autism` — proved by `linarith`
- `softmax_nonneg`, `softmax_sum_one` — proved
- `lse_ge_max` — proved

Proof obligations pending real-analysis scaffolding: `softmax_limit_argmax`, `energy_descent_modern`, `correspondence_limit`.

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Conclusion

The Field-Modulated Hopfield Network resolves the isolation problem at the heart of classical and modern associative memory models. By binding the somatic electromagnetic field to the network's temperature and weight parameters via two coupling equations, the FM-HN provides a biologically grounded mechanism for runtime escape from local minima.

The framework satisfies the Correspondence Principle by construction: under zero somatic stress, both coupling equations vanish and the FM-HN reduces exactly to the standard 2020 Hopfield update. Under high somatic stress, the barriers melt, and the network dynamics enter the quantum tunnelling regime characterised by QUANT-EXP-1.

The three neurodivergent operator modifications (ADHD, ASC, C-PTSD) are not ad hoc additions but emergent properties of the same (β, J, W) parameter space. They are computationally distinct regimes, not clinical labels applied post hoc.

The formal apparatus — field decomposition, correspondence proof, operator characterisation — is type-checked in Lean 4. The empirical apparatus — 3/3 quantum escape vs 0/48 classical — is documented in QUANT-EXP-1.

The limbic layer was not missing from the organism. It was missing from the model.

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